

Where irrigation exists is globally contested

5. Analyses supporting the reply to the reviewers

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1 Preliminary

```
# ANALYSES SUPPORTING THE REPLY TO THE REVIEWERS #####
#####

# This script reproduces every quantitative figure cited in our reply to the
# reviewers. All analyses use the harmonized ensemble at 0.2 deg restricted to
# cropland cells (the same universe as the main analysis).

# Load libraries -----

sensobol::load_packages(c("data.table", "here", "terra", "magrittr"))

# Functions reused from the main analysis -----

source(here("functions", "filter_long_dt_by_crop_mask_aggregated.R"))
source(here("functions", "compute_tau_max_disagreement_fun.R"))

# Harmonized ensemble at 0.2 deg, restricted to cropland -----

dt <- fread(here("datasets", "irrigated_areas_regridded",
                 "irrigated_areas_regridded_02.csv"))
dt[, resolution := "0.2deg"]
crop <- rast(here("irrigated_area_datasets", "GirrE0", "asap_mask_crop_v02.tif"))
dt <- filter_long_dt_by_crop_mask_aggregated(dt = dt, crop_native = crop,
                                             rule = "any")

# Cells x maps -----

wide <- dcast(dt, lon + lat + country + continent ~ dataset, value.var = "mha",
             fill = 0)

maps <- c("gaez_v4", "giam", "gmia", "gripc", "luh2", "meier",
          "mirca2000", "mirca_os", "nagaraj", "spam")
stopifnot(all(maps %in% names(wide)))

# Nominal reference year of each map (Table on temporal characteristics) -----

nominal <- c(giam = 2000, mirca2000 = 2000, gmia = 2005, gripc = 2005,
             nagaraj = 2005, meier = 2005, spam = 2010, mirca_os = 2015,
             gaez_v4 = 2015, luh2 = 2015)

# Mapping families -----

census <- c("gmia", "mirca2000", "mirca_os", "spam", "gaez_v4") # FAO-statistics
rs      <- c("giam", "gripc") # remote sensing
```

```

# Detectability threshold grid (1 ha to 1e5 ha) and 1%-of-cell tolerance -----

taus_ha      <- c(0, 10^(seq(0, 5, by = 0.1)))
taus_mha     <- taus_ha * 1e-6
zero_tol_02  <- 500 # 1% of the 0.2 deg cell, in ha

# Existential disagreement: of the cells where at least one map records
# irrigation (> 0), the fraction in which at least one map records exactly zero.

exist_union <- function(w, cols) {
  A <- as.matrix(w[, ..cols])
  n_present <- rowSums(A > 0, na.rm = TRUE)
  disagree <- n_present > 0 & n_present < length(cols)
  Amin <- do.call(pmin, c(as.data.frame(A), list(na.rm = TRUE)))
  sum(disagree & Amin == 0) / sum(n_present > 0)
}

# Extreme disagreement: fraction of cells whose tau_max exceeds ten times the
# detectability threshold (one map reports > ~10% of the cell, another ~0).

extreme_frac <- function(w, cols, zero_tol = zero_tol_02) {
  tm <- compute_tau_max_existential_fun(w, cols, taus_mha, taus_ha,
                                         zero_tol_ha = zero_tol)
  mean(!is.na(tm$tau_max_ha) & tm$tau_max_ha > 10 * zero_tol)
}

```

2 Reviewer 1

2.1 R1.1: the detectability threshold as a fraction of the grid cell

The same absolute τ (in hectares) is a different fraction of the grid cell at each resolution. We report the equator cell areas and translate key τ values into cell fractions.

```
# Equator cell area (ha) per resolution -----
```

```
R_earth <- 6371
deg <- c("0.2" = 0.2, "0.4" = 0.4, "1" = 1)
cell_ha <- sapply(deg, function(d) (d * pi / 180 * R_earth)^2 * 100)
round(cell_ha)
```

```
##      0.2      0.4      1
##  49457 197829 1236431
```

```
# Key tau values as a percentage of the cell -----
```

```
tau_key <- c(100, 500, 1000, 2500, 5000, 10000, 25000)
tau_tab <- data.table(
  `tau (ha)`      = tau_key,
  `0.2 deg (%)`   = round(100 * tau_key / cell_ha["0.2"], 2),
  `0.4 deg (%)`   = round(100 * tau_key / cell_ha["0.4"], 2),
  `1 deg (%)`     = round(100 * tau_key / cell_ha["1"], 2))
```

```
knitr::kable(tau_tab, caption = "Detectability threshold tau as a percentage of the grid-cell area")
```

Table 1: Detectability threshold tau as a percentage of the grid-cell area at the equator. The analysis centres on tau = 1% of the cell (about 500, 2000 and 12400 ha); extreme disagreement denotes tau_max above ten times this value.

tau (ha)	0.2 deg (%)	0.4 deg (%)	1 deg (%)
100	0.20	0.05	0.01
500	1.01	0.25	0.04
1000	2.02	0.51	0.08
2500	5.05	1.26	0.20
5000	10.11	2.53	0.40
10000	20.22	5.05	0.81
25000	50.55	12.64	2.02

2.2 R1.5: temporal heterogeneity does not explain the disagreement

We test whether the disagreement reflects the different moments the maps represent. Existence of irrigation is the metric throughout (see Preliminary).

```
# Maps that share the 2005 reference year versus the full ensemble -----
```

```
sameyear_2005 <- c("gmia", "gripc", "nagaraj", "meier")
```

```
data.table(
```

```

set = c("Same year (2005)", "Full ensemble"),
maps = c(length(sameyear_2005), length(maps)),
existential_pct = round(100 * c(exist_union(wide, sameyear_2005),
                                exist_union(wide, maps))))

##           set maps existential_pct
##           <char> <int>           <num>
## 1: Same year (2005)      4           69
## 2:   Full ensemble    10           90
# Recompute over every four-map combination to hold ensemble size fixed -----

combs4 <- combn(maps, 4, simplify = FALSE)
ev4    <- sapply(combs4, function(cc) exist_union(wide, cc))
span4  <- sapply(combs4, function(cc) diff(range(nominal[cc])))
same_ev <- exist_union(wide, sameyear_2005)

data.table(
  quantity = c("Minimum", "Median", "Maximum",
               "Same-year (2005) combination", "Most dispersed (1999-2015)"),
  existential_pct = round(100 * c(min(ev4), median(ev4), max(ev4),
                                same_ev, mean(ev4[span4 == max(span4)]))),
  percentile = c(NA, NA, NA, round(100 * mean(ev4 <= same_ev)), NA))

##           quantity existential_pct percentile
##           <char>           <num>           <num>
## 1:           Minimum             15           NA
## 2:           Median             74           NA
## 3:           Maximum             89           NA
## 4: Same-year (2005) combination      69           36
## 5:   Most dispersed (1999-2015)      74           NA
# Direction of disagreement: under real expansion, a recent (2015) map should
# not record zero where an older (2000-2005) map detects irrigation.

old_t    <- c("giam", "mirca2000", "gmia", "gripc") # 2000-2005
recent_t <- c("gaez_v4", "luh2", "mirca_os")        # 2015

pooled_blind <- function(new) {
  A_old <- as.matrix(wide[, ..old_t])
  old_any <- rowSums(A_old > 0, na.rm = TRUE) > 0
  sum(old_any & wide[[new]] == 0, na.rm = TRUE) / sum(old_any)
}

data.table(
  recent_map = c("GAEZ+2015", "LUH2", "MIRCA-OS"),
  sees_zero_where_older_detects_pct = round(100 * sapply(recent_t, pooled_blind)))

##   recent_map sees_zero_where_older_detects_pct

```

##	<char>	<num>
## 1:	GAEZ+2015	19
## 2:	LUH2	83
## 3:	MIRCA-0S	17

3 Reviewer 2

3.1 R2.1: census products agree because they share source data, not reality

The five FAO-statistics products share definitions and much of their source data. We compare their existential disagreement against all five-map subsets.

```
census_ev <- exist_union(wide, census)

combs5 <- combn(maps, 5, simplify = FALSE)
ev5 <- sapply(combs5, function(cc) exist_union(wide, cc))

data.table(
  quantity = c("Census-5 (GMIA, MIRCA2000, MIRCA-OS, SPAM, GAEZ+2015)",
    "All 252 five-map subsets: minimum",
    "All 252 five-map subsets: median",
    "All 252 five-map subsets: maximum"),
  existential_pct = round(100 * c(census_ev, min(ev5), median(ev5), max(ev5))),
  note = c(sprintf("percentile %.0f%%", 100 * mean(ev5 <= census_ev)),
    "", "", ""))
```

```
##                                quantity existential_pct
##                                <char>                <num>
## 1: Census-5 (GMIA, MIRCA2000, MIRCA-OS, SPAM, GAEZ+2015)      24
## 2:                      All 252 five-map subsets: minimum      24
## 3:                      All 252 five-map subsets: median      80
## 4:                      All 252 five-map subsets: maximum      90
##              note
##              <char>
## 1: percentile 0%
## 2:
## 3:
## 4:
```

3.2 R2.2: remote-sensing and census products are not a subset of one another

If remote sensing simply mapped a superset of census irrigation, it would never record zero where a census product detects irrigation. We test both directions and the disagreement within each family.

```
any_pos <- function(cols) rowSums(as.matrix(wide[, ..cols]) > 0, na.rm = TRUE) > 0
cP <- any_pos(census)
rP <- any_pos(rs)

data.table(
  quantity = c("RS records zero where census detects irrigation (% of census-irrigated cells)",
    "Census records zero where RS detects irrigation (% of RS-irrigated cells)",
    "Within RS family: GIAM vs GRIPC existential disagreement (%)",
    "Within census family: five FAO products existential disagreement (%)",
    "Mean global total, census maps (Mha)",
    "Mean global total, RS maps (Mha)"),
```

```

value = c(round(100 * sum(cP & !rP) / sum(cP)),
           round(100 * sum(rP & !cP) / sum(rP)),
           round(100 * exist_union(wide, rs)),
           round(100 * exist_union(wide, census)),
           round(mean(sapply(census, function(m) sum(wide[[m]])))),
           round(mean(sapply(rs, function(m) sum(wide[[m]])))))

```

```

##                                     quantity
##                                     <char>
## 1: RS records zero where census detects irrigation (% of census-irrigated cells)
## 2:   Census records zero where RS detects irrigation (% of RS-irrigated cells)
## 3:                                     Within RS family: GIAM vs GRIPC existential disagreement (%)
## 4:                                     Within census family: five FAO products existential disagreement (%)
## 5:                                     Mean global total, census maps (Mha)
## 6:                                     Mean global total, RS maps (Mha)
##   value
##   <num>
## 1:   36
## 2:   17
## 3:   57
## 4:   24
## 5:  244
## 6:  321

```

*# Per-map global totals underlying the census/RS means (which maps each group
uses follows the reviewer's own classification in his comment) -----*

```

data.table(
  group = c(rep("census", length(census)), rep("remote sensing", length(rs))),
  map = c(census, rs),
  total_mha = round(sapply(c(census, rs), function(m) sum(wide[[m]])), 1))

```

```

##           group      map total_mha
##           <char>   <char>   <num>
## 1:      census    gmia    245.8
## 2:      census mirca2000  210.5
## 3:      census  mirca_os  265.5
## 4:      census    spam    219.6
## 5:      census  gaez_v4   280.3
## 6: remote sensing    giam    335.2
## 7: remote sensing  gripc    306.0

```

3.3 R2.3: census products are compared on a common physical extent

The reviewer notes that the census products map different quantities (area equipped, harvested area, monthly growing area). We do not compare these raw quantities: in `code_original_datasets` we reduce every product to its actual physical irrigated extent, so these magnitude differences do not enter the comparison. The variable used for each product is recorded below.

```

provenance <- data.table(
  product = c("GMIA", "SPAM2010", "MIRCA2000", "MIRCA-OS", "GAEZ+2015", "GIAM"),
  variable_used = c("Actually irrigated area (AEI x AAI fraction), not AEI",
    "Physical irrigated area (SPAM 'A'), capped at cell area",
    "Maximum monthly irrigated area (physical extent), not summed",
    "Maximum monthly growing area (physical extent), not summed",
    "Harvested irrigated area, capped at physical cell area",
    "Net actually irrigated area (TAAI), not annualized (AIA)",
  source = c("code_original_datasets.Rmd L254", "L439, L452", "L341", "L643",
    "L382, L399", "L205"))

knitr::kable(provenance, caption = "Variable extracted for each product (code_original_datasets.Rmd). Every product is reduced to its actual physical irrigated extent, so differences between area equipped, harvested area and monthly growing area do not enter the existence comparison.")

```

Table 2: Variable extracted for each product (code_original_datasets.Rmd). Every product is reduced to its actual physical irrigated extent, so differences between area equipped, harvested area and monthly growing area do not enter the existence comparison.

product	variable_used	source
GMIA	Actually irrigated area (AEI x AAI fraction), not AEI	code_original_datasets.Rmd L254
SPAM2010	Physical irrigated area (SPAM 'A'), capped at cell area	L439, L452
MIRCA2000	Maximum monthly irrigated area (physical extent), not summed	L341
MIRCA-OS	Maximum monthly growing area (physical extent), not summed	L643
GAEZ+2015	Harvested irrigated area, capped at physical cell area	L382, L399
GIAM	Net actually irrigated area (TAAI), not annualized (AIA)	L205

4 Reviewer 3

4.1 R3.2: where the maps agree on the existence of irrigation

The reviewer asks where all datasets agree on irrigation presence. We identify cells where all ten maps detect irrigation above the detectability threshold (1% of the cell, 500 ha at 0.2 deg) – the “agreement on presence” of the tau_max framework – and characterise them, including their overlap with rice production (EarthStat).

```
Apres <- as.matrix(wide[, ..maps])
wide[, agree_presence := rowSums(Apres > zero_tol_02 * 1e-6) == 10]

# Overlap with rice production (EarthStat, regridded to 0.2 deg) -----

rice <- rast(here("datasets", "earthstat_regridded", "res_02",
                 "rice_Production_ll_02.tif"))
wide[, rice_pos := { v <- terra::extract(rice, cbind(lon, lat))[, 1]
                    !is.na(v) & v > 0 }]

ag <- wide[agree_presence == TRUE]

data.table(
  metric = c("Agreement cells (all 10 above 1% of cell)",
             "In Asia (%)",
             "Latitude belt (10th-90th percentile)",
             "Coincide with rice production (%)",
             "All cropland cells coinciding with rice (%)" ),
  value = c(as.character(nrow(ag)),
            as.character(round(100 * mean(ag$continent == "Asia"))),
            sprintf("%.0f-%.0f N", quantile(ag$lat, .10), quantile(ag$lat, .90)),
            as.character(round(100 * mean(ag$rice_pos))),
            as.character(round(100 * mean(wide$rice_pos))))

##           metric      value
##           <char>   <char>
## 1: Agreement cells (all 10 above 1% of cell)      9248
## 2:           In Asia (%)              79
## 3:      Latitude belt (10th-90th percentile) 18-42 N
## 4:           Coincide with rice production (%)       86
## 5: All cropland cells coinciding with rice (%)       56

knitr::kable(ag[, .N, country][order(-N)][1:5],
              caption = "Countries with the most cells where all ten datasets agree irrigation :")
```

Table 3: Countries with the most cells where all ten datasets agree irrigation is present (0.2 deg).

country	N
China	2988
India	1921

country	N
United States	821
Pakistan	376
Iran	229

4.2 R3.14: disagreement does not track irrigation practice

country	cells	existential_pct	extreme_pct
Iran	2931	88	29
Mexico	3126	92	25
United States	17087	93	15

5 Additional analyses: the non-identifiability claim

These analyses support the manuscript section “Irrigated areas are non-identifiable”. The rationale and interpretation are given there and in the SM (Secs. 1.7.3-1.7.5); here we only reproduce the numbers.

5.1 Underdetermination: how many truths are consistent with the data

Determined-present: every map exceeds the strictest admissible threshold ($\min_d A_d > T$).
Determined-absent: every map reports exactly zero (none here by construction, since the universe only contains cells with at least one positive report). All other cells are underdetermined: both classifications are consistent with the ensemble.

```
# UNDERDETERMINATION COUNT #####

A <- as.matrix(wide[, ..maps])
Amin <- do.call(pmin, as.data.frame(A))
Amax <- do.call(pmax, as.data.frame(A))

underdet_fun <- function(T_ha) {
  T_mha <- T_ha * 1e-6
  det_pres <- Amin > T_mha
  und <- !(det_pres | Amax == 0)
  data.table(T_ha = T_ha,
             det_present = sum(det_pres),
             underdetermined = sum(und),
             underdet_pct = round(100 * mean(und), 1),
             existential_pct = round(100 * sum(und & Amin == 0) / sum(und), 1),
             log10_truths = round(sum(und) * log10(2)))
}

knitr::kable(rbindlist(lapply(c(100, 500, 1000), underdet_fun)),
             caption = "Underdetermination of irrigation presence (0.2 deg,
cropland). det\\_present: cells where every map exceeds the
strictest admissible threshold T. existential\\_pct: share of
underdetermined cells where at least one map reports exactly
zero. log10\\_truths: size (in powers of ten) of the set of
complete global presence fields consistent with all ten maps.")
```

Table 5: Underdetermination of irrigation presence (0.2 deg, cropland). `det_present`: cells where every map exceeds the strictest admissible threshold T . `existential_pct`: share of underdetermined cells where at least one map reports exactly zero. `log10_truths`: size (in powers of ten) of the set of complete global presence fields consistent with all ten maps.

<code>T_ha</code>	<code>det_present</code>	<code>underdetermined</code>	<code>underdet_pct</code>	<code>existential_pct</code>	<code>log10_truths</code>
100	11147	118481	91.4	99.0	35666
500	9248	120380	92.9	97.4	36238
1000	7919	121709	93.9	96.4	36638

Consistency checks: determined-present at $T = 500$ ha (9,248 cells) equals the R3.2 agreement-cell count; the existential set (117,294 cells, 90.5%) equals the manuscript's 90% headline.

5.2 Latent-class analysis: testing “identifiable truth plus noise”

K -class latent class models (conditional independence within class) fitted by expectation-maximisation on the 2^{10} response patterns of the binary presence matrix, at $\tau = 0$ and $\tau = 500$ ha, with 50 random starts per fit (Hui and Walter 1980).

K-CLASS LATENT CLASS MODEL VIA EM ON AGGREGATED RESPONSE PATTERNS

```
lca_em <- function(pat, cnt, K, n_start = 50, tol = 1e-8, maxit = 3000,
                  seed = 123) {
  set.seed(seed)
  D <- ncol(pat); n <- sum(cnt)
  best <- NULL; logliks <- numeric(n_start)
  for (s in seq_len(n_start)) {
    x <- rexp(K); pi_k <- x / sum(x)
    rho <- matrix(runif(K * D, .05, .95), K, D)
    ll_old <- -Inf
    for (it in seq_len(maxit)) {
      logp <- sapply(seq_len(K), function(k)
        pat %%% log(rho[k, ]) + (1 - pat) %%% log(1 - rho[k, ]) + log(pi_k[k]))
      m <- apply(logp, 1, max)
      lse <- m + log(rowSums(exp(logp - m)))
      ll <- sum(cnt * lse)
      post <- exp(logp - lse)
      w <- post * cnt
      Nk <- colSums(w)
      pi_k <- Nk / n
      rho <- pmin(pmax(crossprod(w, pat) / Nk, 1e-6), 1 - 1e-6)
      if (abs(ll - ll_old) < tol) break
      ll_old <- ll
    }
    logliks[s] <- ll
    if (is.null(best) || ll > best$ll)
      best <- list(ll = ll, pi = pi_k, rho = rho, post = post)
  }
  npar <- (K - 1) + K * D
  c(best, list(npar = npar, bic = -2 * best$ll + npar * log(n),
              logliks = logliks))
}
```

Fit K = 1-4, report fit, error rates, residual dependence, 3-class profile --

```
lca_report_fun <- function(Y, label) {
  D <- ncol(Y); n <- nrow(Y)
  code <- as.integer(Y %%% 2^(0:(D - 1)))
```

```

tab <- data.table(code = code)[, .N, by = code][order(code)]
pat <- t(sapply(tab$code, function(z) as.integer(intToBits(z)[1:D])))
cnt <- tab$N
fits <- lapply(1:4, function(K) lca_em(pat, cnt, K))

comp <- data.table(K = 1:4,
  loglik = sapply(fits, function(f) round(f$ll, 1)),
  npar    = sapply(fits, function(f) f$npar),
  BIC     = sapply(fits, function(f) round(f$bic, 1)))

f2 <- fits[[2]]
irr <- which.max(rowMeans(f2$rho)); no <- setdiff(1:2, irr)
err <- data.table(map = maps,
  FNR = round(1 - f2$rho[irr, ], 3),
  FPR = round(f2$rho[no, ], 3))[order(-FNR)]

logp <- sapply(1:2, function(k)
  pat %>% log(f2$rho[k, ]) + (1 - pat) %>% log(1 - f2$rho[k, ]) +
  log(f2$pi[k]))
m <- apply(logp, 1, max)
e <- n * exp(m + log(rowSums(exp(logp - m))))
G2 <- 2 * sum(cnt * log(cnt / e))
df <- nrow(pat) - 1 - f2$npar

P_obs <- crossprod(Y) / n
P_exp <- f2$pi[1] * tcrossprod(f2$rho[1, ]) +
  f2$pi[2] * tcrossprod(f2$rho[2, ])
R <- (P_obs - P_exp) * 100
ut <- which(upper.tri(R), arr.ind = TRUE)
res <- data.table(m1 = maps[ut[, 1]], m2 = maps[ut[, 2]],
  resid = round(R[ut, 2])[order(-abs(resid))][1:5])

f3 <- fits[[3]]
ord <- order(rowMeans(f3$rho))
pr <- data.table(class = sprintf("pi = %.2f", f3$pi[ord]),
  round(as.data.table(f3$rho[ord, , drop = FALSE]), 2))
setnames(pr, c("class", maps))
post3 <- f3$post[match(code, tab$code), ]
mid_share <- 100 * mean(apply(post3, 1, which.max) == ord[2])
ll_r <- round(f2$logliks, 1)

cat("\n####", label, "----", n, "cells,", nrow(pat), "patterns\n")
print(comp)
cat(sprintf("\n2-class fit: G2 = %.0f on df = %d (G2/df = %.1f)\n",
  G2, df, G2 / df))
cat(sprintf("2-class multistart: %d distinct loglik modes in 50 starts\n",
  length(unique(ll_r))))

```

```

cat("\nImplied per-map error rates (2-class):\n"); print(err)
cat("\nLargest residual pairwise associations (obs - exp P(both = 1), x100):\n")
print(res)
cat(sprintf("\n3-class model: BIC improvement over 2-class = %.0f;",
            fits[[2]]$bic - f3$bic))
cat(sprintf(" middle-class posterior share = %.1f%\n", mid_share))
cat("3-class item-probability profiles (rows ordered low to high):\n")
print(pr)
invisible(fits)
}

```

```

# RUN AT BOTH DETECTION CONVENTIONS #####

```

```

fits_tau0 <- lca_report_fun(1L * (A > 0), "tau = 0 (any positive area)")

```

```

##
## ##### tau = 0 (any positive area) ---- 129628 cells, 462 patterns
##      K      loglik  npar      BIC
##      <int>      <num> <num>      <num>
## 1:      1 -746739.4      10 1493597
## 2:      2 -580325.3      21 1160898
## 3:      3 -522411.1      32 1045199
## 4:      4 -513810.4      43 1028127
##
## 2-class fit: G2 = 165999 on df = 440 (G2/df = 377.3)
## 2-class multistart: 1 distinct loglik modes in 50 starts
##
## Implied per-map error rates (2-class):
##      map      FNR      FPR
##      <char> <num> <num>
## 1:      luh2 0.802 0.041
## 2:      giam 0.546 0.381
## 3:      nagaraj 0.504 0.085
## 4:      gripc 0.480 0.528
## 5:      meier 0.329 0.187
## 6:      mirca2000 0.113 0.246
## 7:      spam 0.088 0.036
## 8:      mirca_os 0.018 0.106
## 9:      gaez_v4 0.013 0.006
## 10:      gmia 0.004 0.139
##
## Largest residual pairwise associations (obs - exp P(both = 1), x100):
##      m1      m2 resid
##      <char> <char> <num>
## 1:      meier nagaraj 11.55
## 2:      gripc nagaraj  9.65
## 3:      gripc  meier  7.29
## 4:      giam nagaraj  7.04

```

```
## 5:   giam   gripc  6.05
##
## 3-class model: BIC improvement over 2-class = 115699; middle-class posterior share = 43.3%
## 3-class item-probability profiles (rows ordered low to high):
##      class gaez_v4  giam  gmia gripc  luh2 meier mirca2000 mirca_os nagaraj
##      <char>   <num> <num> <num> <num> <num> <num>      <num>   <num>   <num>
## 1: pi = 0.17    0.00  0.41  0.03  0.57  0.04  0.20      0.20    0.00    0.09
## 2: pi = 0.44    0.93  0.25  0.99  0.24  0.02  0.35      0.82    0.97    0.10
## 3: pi = 0.39    1.00  0.67  1.00  0.82  0.38  0.99      0.94    0.99    0.91
##      spam
##      <num>
## 1:  0.04
## 2:  0.81
## 3:  0.97
```

```
fits_tau500 <- lca_report_fun(1L * (A > 5e-4), "tau = 500 ha (1% of cell)")
```

```
##
## ##### tau = 500 ha (1% of cell) ---- 129628 cells, 854 patterns
##      K      loglik  npar      BIC
##      <int>      <num> <num>      <num>
## 1:      1 -802738.4    10 1605594.6
## 2:      2 -481300.7    21  962848.5
## 3:      3 -458060.3    32  916497.3
## 4:      4 -452557.5    43  905621.2
##
## 2-class fit: G2 = 68582 on df = 832 (G2/df = 82.4)
## 2-class multistart: 1 distinct loglik modes in 50 starts
##
## Implied per-map error rates (2-class):
##      map   FNR   FPR
##      <char> <num> <num>
## 1:      luh2 0.619 0.031
## 2:      giam 0.449 0.175
## 3:      gripc 0.398 0.131
## 4:      spam 0.223 0.015
## 5: mirca2000 0.182 0.040
## 6:   nagaraj 0.116 0.153
## 7: mirca_os 0.089 0.032
## 8:   gaez_v4 0.076 0.045
## 9:      gmia 0.066 0.010
## 10:   meier 0.028 0.100
##
## Largest residual pairwise associations (obs - exp P(both = 1), x100):
##      m1      m2 resid
##      <char>   <char> <num>
## 1:      giam      gripc  4.68
## 2: mirca2000      spam  2.36
```

```
## 3:      giam mirca2000  2.08
## 4:      gripc  nagaraj  2.06
## 5:      gripc    luh2   2.05
##
## 3-class model: BIC improvement over 2-class = 46351; middle-class posterior share = 15.5%
## 3-class item-probability profiles (rows ordered low to high):
##      class gaez_v4  giam  gmia gripc  luh2 meier mirca2000 mirca_os nagaraj
##      <char>  <num> <num> <num> <num> <num> <num>      <num>      <num>      <num>
## 1: pi = 0.60    0.02  0.17  0.00  0.13  0.03  0.07      0.03      0.01      0.13
## 2: pi = 0.16    0.76  0.24  0.66  0.27  0.14  0.86      0.45      0.66      0.68
## 3: pi = 0.24    0.97  0.71  1.00  0.76  0.50  1.00      0.97      0.99      0.96
##      spam
##      <num>
## 1:  0.01
## 2:  0.35
## 3:  0.96
```

5.3 The best-products test over the United States

LANID (Xie and Lark 2021; 30 m, Landsat, annual), MirAD-US v4 (Pervez and Brown 2010; 250 m, MODIS anchored to USDA county statistics) and AIM-HPA (Deines et al. 2019; 30 m, Landsat, High Plains Aquifer) compared in 2012 and 2017, the two years all three cover. Sources: LANID doi:10.5281/zenodo.5548555; MirAD-US doi:10.5066/P9NA3EO8; AIM-HPA doi:10.4211/hs.a371fd69d41b4232806d81e17fe4efcb. The aggregation to the 0.2° grid (warp-sum of binary irrigated pixels × pixel area: 0.09 ha at 30 m, 6.25 ha at 250 m) is documented in the non-evaluated chunk below; the aggregated table ships with the repository (us_regional_products/). CONUS totals as a sanity check: LANID 23.1/24.2 Mha, MirAD 22.5/23.8 Mha (2012/2017).

```
# AGGREGATION OF THE US REGIONAL PRODUCTS (documentation; not run) #####

template <- rast(xmin = -126, xmax = -66, ymin = 24, ymax = 50,
                resolution = 0.2, crs = "EPSG:4326")

agg_to_ha <- function(f, irr_val, px_ha) {
  ir <- (rast(f) == irr_val) * 1.0          # binary irrigated
  project(ir, template, method = "sum") * px_ha # pixel count x pixel area
}

# lanid2012/2017.tif: 30 m, CONUS Albers, 0/1 (Zenodo record 5548555)
# mirad250_12/17v4.tif: 250 m, Lambert azimuthal EA, 0/1 (ScienceBase)
# {2012,2017}_AIM-HPA_rse_finalMapBinary.tif: 30 m; 0 = NoData, 1 = irrigated,
#   2 = not irrigated (HydroShare); validity = fraction of cell with value > 0

# EXISTENTIAL DISAGREEMENT: REGIONAL PRODUCTS VS THE GLOBAL ENSEMBLE #####

# The CSV holds the product values extracted at the paper's 0.2 deg cell
# centres for every US cropland cell, so it joins back on lon/lat exactly
# (rounded to one decimal to be robust to floating-point noise).
```

```

reg <- fread(here("us_regional_products", "us_regional_products_02.csv"))
reg[, `:=`(lon = round(lon, 1), lat = round(lat, 1))]

us <- copy(wide[country == "United States"])
us[, `:=`(lon = round(lon, 1), lat = round(lat, 1))]
us <- merge(us, reg, by = c("lon", "lat"))
stopifnot(nrow(us) == nrow(reg))

# Existential: of union cells (>0 somewhere), % with at least one exact zero.
# Strong: % of union cells where one product is 0 and another exceeds 1% of
# the cell. Regional columns are in ha; the global maps are in Mha.

exist_pct <- function(d, cols) {
  A <- as.matrix(d[, ..cols])
  np <- rowSums(A > 0); un <- np > 0
  round(100 * sum(un & np < length(cols)) / sum(un), 1)
}

strong_pct <- function(d, cols, big) {
  A <- as.matrix(d[, ..cols])
  un <- rowSums(A > 0) > 0
  round(100 * sum(un & apply(A, 1, min) == 0 &
    apply(A, 1, max) > big) / sum(un), 1)
}

dir_pct <- function sumun_____

```

```
knitr::kable(out, caption = "Existential disagreement (of the cells where at
least one product records irrigation, the percentage where at
least one records exactly zero) and strong contradictions (one
product records zero, another exceeds 1\\% of the cell) among the
best US regional irrigation products, benchmarked against the ten
global maps over the same cells. CONUS: 17,087 US cropland cells;
High Plains: 1,549 cells within the AIM-HPA footprint.")
```

Table 6: Existential disagreement (of the cells where at least one product records irrigation, the percentage where at least one records exactly zero) and strong contradictions (one product records zero, another exceeds 1% of the cell) among the best US regional irrigation products, benchmarked against the ten global maps over the same cells. CONUS: 17,087 US cropland cells; High Plains: 1,549 cells within the AIM-HPA footprint.

universe	comparison	existential	strong
CONUS	LANID x MlrAD 2012	32.1	1.3
CONUS	LANID x MlrAD 2017	33.6	3.0
CONUS	Ten global maps	93.0	46.6
High Plains	LANID x MlrAD 2012	5.7	0.0
High Plains	LANID x MlrAD 2017	5.3	0.2
High Plains	LANID x MlrAD x AIM 2012	7.8	0.3
High Plains	LANID x MlrAD x AIM 2017	8.0	0.7
High Plains	Ten global maps	75.0	70.7

```
# Directional asymmetry (mirrors the global census-vs-remote-sensing axis) -----
```

```
data.table(
  year = c(2012, 2017),
  lanid0_mirad_pos = c(dir_pct(us, "lanid12", "mirad12"),
                        dir_pct(us, "lanid17", "mirad17")),
  mirad0_lanid_pos = c(dir_pct(us, "mirad12", "lanid12"),
                        dir_pct(us, "mirad17", "lanid17")))
```

```
##      year lanid0_mirad_pos mirad0_lanid_pos
##      <num>          <num>          <num>
## 1:  2012           23.1           9.0
## 2:  2017           26.0           7.6
```

```
# GEOGRAPHIC AND MAGNITUDE STRUCTURE OF THE LANID-MIRAD DISAGREEMENT #####
```

```
# Directional cells by longitude (100W = the classic humid/arid divide) and
# the median irrigated area at stake in each direction.
```

```
struct <- rbindlist(lapply(c("12", "17"), function(yr) {
  L <- us[[paste0("lanid", yr)]]; M <- us[[paste0("mirad", yr)]]
  un <- L > 0 | M > 0
  lm <- un & L == 0 & M > 0; ml <- un & M == 0 & L > 0; both <- L > 0 & M > 0
  data.table(year = paste0("20", yr),
```

```

        direction = c("MirAD > 0, LANID = 0", "LANID > 0, MirAD = 0",
                      "Both > 0"),
        cells = c(sum(lm), sum(ml), sum(both)),
        east_100W_pct = round(100 * c(mean(us$lon[lm] > -100),
                                     mean(us$lon[ml] > -100),
                                     mean(us$lon[both] > -100))),
        median_ha = round(c(median(M[lm]), median(L[ml]),
                             median(pmin(L, M)[both]))))
    )))
struct

```

```

##      year      direction cells east_100W_pct median_ha
##      <char>      <char> <int>      <num>      <num>
## 1:  2012 MirAD > 0, LANID = 0  3296          93         50
## 2:  2012 LANID > 0, MirAD = 0  1292          33         18
## 3:  2012      Both > 0  9693          52        250
## 4:  2017 MirAD > 0, LANID = 0  3812          93         75
## 5:  2017 LANID > 0, MirAD = 0  1106          26         17
## 6:  2017      Both > 0  9731          51        272

```

```

cat("Baseline: union cells east of 100W:",
    round(100 * mean(us$lon[us$lanid12 > 0 | us$mirad12 > 0] > -100)), "%\n")

```

```
## Baseline: union cells east of 100W: 60 %
```

```
# Pairwise existential disagreement within the High Plains -----
```

```

pair_exist <- function(a, b) {
  un <- a > 0 | b > 0
  round(100 * mean((a == 0 | b == 0)[un]), 1)
}
for (yr in c("12", "17"))
  cat(sprintf(
    "HPA 20%s pairwise existential: LANIDxMirAD %.1f | LANIDxAIM %.1f | MirADxAIM %.1f\n",
    yr,
    pair_exist(hpa[[paste0("lanid", yr)]], hpa[[paste0("mirad", yr)]]),
    pair_exist(hpa[[paste0("lanid", yr)]], hpa[[paste0("aim", yr)]]),
    pair_exist(hpa[[paste0("mirad", yr)]], hpa[[paste0("aim", yr)]])
  ))

```

```
## HPA 2012 pairwise existential: LANIDxMirAD 5.7 | LANIDxAIM 3.6 | MirADxAIM 6.5
```

```
## HPA 2017 pairwise existential: LANIDxMirAD 5.3 | LANIDxAIM 4.4 | MirADxAIM 6.6
```

The interpretation of these results is given in the manuscript (section “Irrigated areas are non-identifiable”) and the SM (Sec. 1.7.5).

6 Session information

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] C
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] magrittr_2.0.5   terra_1.8-93     here_1.0.2      data.table_1.18.4
##
## loaded via a namespace (and not attached):
## [1] vctrs_0.7.3      cli_3.6.6        knitr_1.51       rlang_1.2.0
## [5] xfun_0.55        otel_0.2.0       generics_0.1.4   S7_0.2.2
## [9] glue_1.8.1       rprojroot_2.1.1  htmltools_0.5.9  scales_1.4.0
## [13] sensobol_1.2.0   rmarkdown_2.30   grid_4.5.2       tibble_3.3.1
## [17] evaluate_1.0.5   fastmap_1.2.0    yaml_2.3.12      lifecycle_1.0.5
## [21] compiler_4.5.2   codetools_0.2-20 dplyr_1.1.4      RColorBrewer_1.1-3
## [25] Rcpp_1.1.1-1.1   pkgconfig_2.0.3  farver_2.1.2     digest_0.6.39
## [29] R6_2.6.1         tidyselect_1.2.1 Rdpack_2.6.6     pillar_1.11.1
## [33] rbibutils_2.4.1  tools_4.5.2      gtable_0.3.6     ggplot2_4.0.3.9000
```